Wanghan Xu

Curriculum Vitae

September 14, 2001
Shanghai, China
Shanghai Jiao Tong University
Stanford University
(+86)13588582062
xu_wanghan@sjtu.edu.cn
Homepage
Google Scholar
GitHub

Education

2026.05-Present **Visiting Scholar, Stanford University**. Research Topics: Embodied AI, Automated Science Discovery from Virtual World to Physical World. Supervised by Prof. Le Cong**.**

2024.09-Present **PhD**, Information and Communication Technology, Shanghai Jiao Tong University. Research Topics: LLMs, Agents, AI Scientists, Generative AI. Supervised by Prof. Wanli Ouyang.

2020.09-2024.06 Bachelor of Computer Science and Technology, Xi'an Jiaotong University. GPA: **100.01**(94.01+6)/100, Rank: 1/196**,** Awarded as **Top Ten Outstanding Student**.

Key Contributor Publications

2026 **Wanghan Xu***, Shuo Li*, Tianlin Ye, Qinglong Cao, Yixin Chen, Hengjian Gao ... (50 authors). *ResearchClawBench: A Benchmark for End-to-End Autonomous Scientific Research* Link

2026 **Wanghan Xu**, Yuhao Zhou, Hengyuan Zhao, Shuo Li, Dianzhi Yu, Zhenfei Yin, Yaowen Hu, Fengli Xu, Wanli Ouyang, Wenlong Zhang, Lei Bai. *ReCrit: Transition-Aware Reinforcement Learning for Scientific Critic Reasoning* Link

2025 **Wanghan Xu**, Yuhao Zhou, Yifan Zhou, Qinglong Cao, Shuo Li, Jia Bu ... (107 authors). *Probing Scientific General Intelligence of LLMs with Scientist-Aligned Workflows* Link

2025 Tao Han, **Wanghan Xu**, Junchao Gong, Xiaoyu Yue, Song Guo, Luping Zhou, Lei Bai. *InfGen: A Resolution-Agnostic Paradigm for Scalable Image Synthesis* **(ICCV 2025 Accept)** Link

2025 Xiangru Tang*, **Wanghan Xu***, Yujie Wang*, Zijie Guo*, Daniel Shao, Jiapeng Chen, Cixuan Zhang, Ziyi Wang, Lixin Zhang, Guancheng Wan, Wenlong Zhang, Lei Bai, Zhenfei Yin, Philip Torr, Hanrui Wang, Di Jin. *Eigen-Agent: Adaptive Multi-Agent Refinement with Monitor-Based RAG for Scientific Reasoning* **(ICLR 2026 Accept)** Link

2025 Ming Hu, Chenglong Ma, Wei Li, **Wanghan Xu**, Jiamin Wu, Jucheng Hu, Tianbin Li, Guohang Zhuang, Jiaqi Liu …(103 authors). *A survey of scientific large language models: From data foundations to agent frontiers* Link

2025 **Wanghan Xu**, Wenlong Zhang, Fenghua Ling, Ben Fei, Yusong Hu, Runmin Ma, Bo Zhang, Fangxuan Ren, Jintai Lin, Wanli Ouyang, Lei Bai. *Manalyzer: End-to-end Automated Meta-analysis with Multi-agent System* Link

2025 **Wanghan Xu**, Xiaoyu Yue, Zidong Wang, Yao Teng, Wenlong Zhang, Xihui Liu, Luping Zhou, Wanli Ouyang, Lei Bai. *Exploring Representation-Aligned Latent Space for Better Generation* Link

2025 **Wanghan Xu**, Xiangyu Zhao, Yuhao Zhou, Xiaoyu Yue, Ben Fei, Fenghua Ling, Wenlong Zhang, Lei Bai. *EarthSE: A Benchmark Evaluating Earth Scientific Exploration Capability for Large Language Models* **(ICLR 2026 Accept)** Link

2025 Xiangyu Zhao*, **Wanghan Xu***, Bo Liu, Yuhao Zhou, Fenghua Ling, Ben Fei, Xiaoyu Yue, Lei Bai, Wenlong Zhang, Xiao-Ming Wu. *MSEarth: A Benchmark for Multimodal Scientific Comprehension of Earth Science* **(ACL Main Conference 2026 Accept)** Link

2024 **Wanghan Xu**, Fenghua Ling, Wenlong Zhang, Tao Han, Hao Chen, Wanli Ouyang, Lei Bai. *Generalizing Weather Forecast to Fine-grained Temporal Scales via Physics-AI Hybrid Modeling* **(NeurIPS 2024 Accept)** Link

2024 **Wanghan Xu**, Kang Chen, Tao Han, Hao Chen, Wanli Ouyang, Lei Bai. *ExtremeCast: Boosting Extreme Value Prediction for Global Weather Forecast* Link

2024 Junchao Gong, Lei Bai, Peng Ye, **Wanghan Xu**, Na Liu, Jianhua Dai, Xiaokang Yang, Wanli Ouyang. *CasCast: Skillful High-resolution Precipitation Nowcasting via Cascaded Modelling* **(ICML 2024 Accept)** Link

2023 **Wanghan Xu**. *Is 3-(F)WL Enough to Distinguish All 3D Graphs?* Link

2023 **Wanghan Xu**, Bin Shi, Ao Liu, Jiqiang Zhang, Bo Dong. *GraphPub: Generation of Differential Privacy Graph with High Availability* Link

2023 **Wanghan Xu**, Bin Shi, Jiqiang Zhang, Zhiyuan Feng, Tianze Pan, Bo Dong. *MDP: Privacy-Preserving GNN Based on Matrix Decomposition and Differential Privacy* **(IEEE JCC 2023 Accept)** Link

Collaborative Publications

2026 *Dynamic Mixture of Latent Memories for Self-Evolving Agents* Link

2026 *Sci-PRM: A Tool Aware Process Reward Model for Scientific Reasoning Verification* **(KDD 2026 Accept)** Link

2026 *Intern-S1-Pro: Scientific Multimodal Foundation Model at Trillion Scale* Link

2026 *InternAgent-1.5: A Unified Agentic Framework for Long-Horizon Autonomous Scientific Discovery* Link

2026 *Earth Science Foundation Models: From Perception to Reasoning and Discovery* Link

2025 *Intern-S1: A Scientific Multimodal Foundation Model* Link

2025 *SciEvalKit: An Open-source Evaluation Toolkit for Scientific General Intelligence* Link

2025 *Earth-Agent: Unlocking the Full Landscape of Earth Observation with Agents* **(ICLR 2026 Accept)** Link

2025 *Omni-Weather: Unified Multimodal Foundation Model for Weather Generation and Understanding* **(ICLR 2026 Accept)** Link

2025 *EarthLink: A Self-Evolving AI Agent for Climate Science* Link

2025 *Align-DA: Align Score-based Atmospheric Data Assimilation with Multiple Preferences* **(NeurIPS 2025 Accept)** Link

2025 *DAWP: A Framework for Global Observation Forecasting via Data Assimilation and Weather Prediction in Satellite Observation Space* **(NeurIPS 2025 Accept)** Link

2025 *LO-SDA: Latent Optimization for Score-based Atmospheric Data Assimilation* Link

2024 *WEATHER-5K: A Large-scale Global Station Weather Dataset Towards Comprehensive Time-series Forecasting Benchmark* Link

2024 *CRA5: Extreme Compression of ERA5 for Portable Global Climate and Weather Research via an Efficient Variational Transformer* Link

Selected Ongoing Projects

2026 Ongoing *Towards Self-Evolving Scientific Discovery with Online Policy Distillation*

Abstract: Designed a self-evolving scientific discovery agent system with multi-round online self-distillation, where Challenger constructs scientific environments, Solver performs full research workflows (literature review, coding, reporting, etc.), and Judge supervises both.

Internship Experience

2023.07- Present **Shanghai AI Lab. Shanghai, China.**
Research Topics: AI for Science, Physics-informed ML, Generation

2022.03- 2023.06 **Shaanxi Provincial Key Laboratory of Tiandi Network Technology.**
Research Topics: Graph Neural Networks

Awards & Honors

2023 Top Ten Outstanding Student Models of Xi'an Jiaotong University
2023 American Mathematical Contest in Modeling (MCM/ICM) Finalist
2023 China Scientist Scholarship (20,000 ¥)
2022 National Scholarship
2022 RoboCup Robot World Cup China First Prize
2021 National Mathematical Contest in Modeling Shaanxi First Prize
2021 National College Mathematics Competition Shaanxi Province First Prize
2021 Xi'an Jiaotong University First Prize Scholarship

Recommendation Letter

2025 Prof. Wanli Ouyang, Professor at the Chinese University of Hong Kong, Leading Scientist (领军科学家) at the Shanghai Artificial Intelligence Laboratory. Link

2023 Prof. Qinghua Zheng, Academician of the Chinese Academy of Engineering (中国工程院院士), President of Tongji University. Link